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MFR-YOLO: advancing UAV object detection with multi-scale feature refinement via deformable convolution and global attention

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Unmanned aerial vehicle imagery exhibits extreme scale variance, high small-object density, and arbitrary geometric distortions, which continue to pose challenges to the state-of-the-art detectors like YOLO series. This paper proposes MFR-YOLO, an enhanced model built upon a multi-scale feature refinement network. First, a multi-scale feature extraction module is designed, integrating SPD-Conv to preserve fine-grained details and employing DCNv4 to dynamically adjust receptive fields for adapting to objects of varying scales and orientations. Standard convolutions in the backbone are systematically replaced with DCNv4 layers, enhancing feature propagation from low-level edges to high-level semantics. A lightweight Global Attention Module is incorporated into both the backbone and neck, leveraging dual channel-spatial self-attention to suppress background clutter and amplify critical object features. Additionally, the original C3K2 bottleneck is upgraded with a Pyramid-Pooling Attention mechanism, enriching multi-scale representations by fusing local and global contextual information. Comprehensive evaluations on the VisDrone2021 and UA-DETRAC benchmarks demonstrate that MFR-YOLO achieves superior overall detection accuracy, exhibits significant gains in small object detection, and maintains a balance between accuracy and real-time efficiency. Ablation studies validate the contribution of each proposed component. This research provides a novel technical framework for object detection algorithms in complex scenes.

Keywords Object detection, YOLO, Multi-scale feature extraction, Global attention, Deformable convolution

In recent years, with the rapid advancement of unmanned aerial vehicle (UAV) technology, its applications have expanded significantly across various fields such as agricultural monitoring, traffic surveillance, emergency rescue, and urban security[1]. As a core technology for UAV environmental perception, object detection directly determines the accuracy and efficiency of task execution[2]. From the perspective of technological evolution, the improvement in the performance of UAV-borne sensors has made it possible to acquire massive visual data in real-time[3]. However, the challenge of rapidly and accurately extracting target information from this data remains a critical bottleneck constraining intelligent UAV applications. Particularly in complex and dynamic scenarios, such as traffic flow monitoring on highways, debris search and rescue after natural disasters, and crowd control at large-scale events, extremely demanding requirements are placed on the robustness and real-time performance of object detection[4].

However, the object detection scenarios from the UAV perspective have significant particularities, posing multiple challenges to existing algorithms[5]. Firstly, constrained by the UAV's flight altitude, ground targets often occupy only 10–30 pixels in images, with extremely sparse feature information and a high proportion of small-scale targets, leading to frequent missed detections in traditional detection models[6]. Secondly, tilted flight or attitude variations of UAVs can induce nonlinear geometric distortions of targets, yet networks with fixed convolution kernels struggle to adapt to such dynamic changes[7]. Thirdly, background elements such as the sky, vegetation, and buildings often overlap with target features, making the model prone to misclassifying background as the target[8]. Fourthly, in dynamic UAV flight scenarios, target positions change rapidly over time,

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and the detection frame rate must be maintained above 25 FPS to meet the requirements of real-time decision-making, which places higher demands on the lightweight design of the model[9]. Moreover, overly complex models may enhance accuracy but cause decision delays due to computational latency, while excessively lightweight models struggle with scene complexity[10]. This further exacerbates computational challenges for UAV platforms.

The YOLO series algorithms have been widely applied in the field of object detection by virtue of their core advantage of balancing efficiency and real-time performance[11]. Among them, YOLOv12 demonstrates outstanding performance in general object detection scenarios through its efficient single-stage detection architecture and lightweight design[12]. However, it still has obvious shortcomings in UAV scenarios: its backbone network employs traditional convolutions combined with stride convolutions and pooling operations for feature map compression, which can easily lead to the loss of fine-grained features of small objects. The feature fusion mechanism lacks targeted adaptation for perspective-distorted objects, making it difficult to effectively capture complete feature representations of distorted targets. Furthermore, the model lacks a dedicated background suppression module, resulting in significantly reduced detection robustness under the strong interference of complex backgrounds in drone scenarios[13]. Although in recent years, researchers have attempted to optimize the model by introducing attention mechanisms and improving feature fusion, most methods do not fully consider the multi-dimensional constraints of UAV scenarios, making it difficult to balance accuracy and efficiency[14]. For example, some models increase the computational load significantly by stacking deep networks to improve the small-target detection ability[15]. Some methods only optimize for a single problem and fail to form a systematic solution[16]. Recent transformer-based approaches[17,18] have shown promise in addressing small object sparsity and feature ambiguity in high-resolution imagery, but they rely on complex transformer structures, which are not suitable for UAV scenarios with strict real-time requirements.

Therefore, in response to the unique requirements of UAV scenarios, customizing and improving YOLOv12 to enhance its performance in small-target detection, distortion adaptation, and background anti-interference has important theoretical value and application significance. In this study, an enhanced YOLOv12 model suitable for UAV scenarios is constructed by designing a multi-scale feature-extraction module, introducing deformable convolutions to enhance feature transfer, embedding a global attention mechanism to suppress background interference, and optimizing the feature-fusion module. The results of comparative experiments on datasets such as VisDrone and UA-DETRAC show that the proposed improved model significantly outperforms the YOLOv12 baseline method and other mainstream detection models in terms of detection accuracy and missed-detection rate. Based on the above research, the main contributions of this paper can be summarized as follows:

- The multi-scale feature extraction module (MSFEM) was constructed by innovatively fusing SPD-Conv (Spatial Pyramid Depthwise Convolution) and DCNv4 (Deformable Convolution v4) to solve the problem of small target degradation and geometric distortion. Without increasing the number of parameters, the module can not only completely retain high-resolution detail information, but also dynamically adjust the receptive field to adapt to the feature extraction requirements of different scale targets, so as to realize the dual optimization of detail preservation and deformation modeling.
- DCNv4 is used to enhance and optimize the backbone network, and the feature transfer efficiency and distortion adaptation ability of the network are greatly improved by replacing the traditional convolution operation, so that the feature expression is more suitable for the target shape changes in complex scenes, which lays the foundation for subsequent accurate recognition.
- A dual-path Global Attention Module (GAM) is embedded in the backbone network and neck structure for the first time. Leveraging the channel-spatial dual-dimensional self-attention mechanism, it effectively suppresses background noise interference, strengthens the feature response of key target regions, significantly improves target saliency, and helps the model accurately focus on core recognition objects.
- A C3K2-PPA module is constructed by integrating pyramid pooling attention. Through a design featuring local-global cascaded branches, it efficiently aggregates multi-scale contextual information, breaking the limitations of single-scale features and further enriching the hierarchical and comprehensive nature of target features.

The structure of this paper is as follows: "Related work" reviews the related work. "Methodology" introduces the proposed method in detail. "Experiments" presents the experimental settings and result analysis. "Conclusion" conducts a summary and discussion.

Related work

Applications of UAV visual object detection

With the maturation of UAV manufacturing technology, flight control systems, and sensor technology, UAV, as efficient and flexible low-altitude data-acquisition platforms, have expanded their application boundaries from traditional aerial photography and military reconnaissance to multiple important sectors of the national economy, giving rise to a large number of intelligent solutions based on visual object detection[19].

In the field of agricultural monitoring, as an important support for precision agriculture, UAV equipped with high-resolution cameras can monitor the growth status of crops, pest and disease conditions, and irrigation needs in real-time[20]. By combining with object-detection algorithms, they can achieve automatic identification and counting of specific crops, locate weeds, and accurately identify disease-affected areas[21]. In the field of traffic surveillance, with their wide field of view and flexible mobility, UAV can fill the monitoring blind spots of fixed cameras in intelligent transportation systems. They can not only monitor road traffic flow in real-time, detect traffic violations, identify traffic accidents, but also conduct traffic flow statistics and analysis[22]. In the field of emergency rescue, for natural disasters such as earthquakes, floods, and fires, UAV can quickly enter dangerous areas to carry out search and rescue missions. By being equipped with thermal-imaging or visible-light cameras

and combined with object-detection algorithms, they can rapidly locate trapped people, vehicles, or rescue supplies in ruins or complex terrains[23]. In the field of urban security, as an important complement to smart city construction, UAV inspections are widely used in scenarios such as crowd density monitoring and abnormal behavior recognition at large-scale events, inspection and fault detection of key infrastructure, and public-safety crime tracking. These applications place extremely high demands on the real-time performance, robustness, and multi-class recognition ability of object-detection algorithms[24]. Although significant progress has been made in applications across various fields, the unique aerial-photography perspective of UAV also brings common technical challenges such as drastic target-scale changes, dense small targets, complex backgrounds, variable shooting angles, and limited platform computing resources[25].

YOLO series object detection models

Since the advent of the YOLO series algorithms, they have established a pivotal position in single-stage object detection through their exceptional balance between inference speed and detection accuracy. The development trajectory of this series clearly demonstrates the evolutionary path of object detection technology, which shifts from an early focus on accuracy improvement to a gradual emphasis on balancing efficiency and deployment practicality[26]. Looking back at its development, the pioneering works such as YOLOv1 to v3 proposed by Redmon et al.[27] have laid a solid foundation for this research direction. Among these, YOLOv1 served as the starting point, pioneering the formulation of object detection as a unified regression problem. By using an end-to-end neural network to directly predict bounding boxes and class probabilities, it not only drastically simplified the detection process but also formally inaugurated a new paradigm for real-time object detection. Subsequently, YOLOv2 introduced the anchor box mechanism and multi-scale training strategy[28], while YOLOv3 significantly improved the performance of small target detection through the Darknet-53 backbone network and a multi-scale prediction mechanism similar to FPN[29]. The performance optimization and engineering exploration stage YOLOv4-V8 further integrates advanced technologies, such as the "Bag of Freebies/Specials" strategy of YOLOv4[30], the fusion of CSPNet and PANet features, and the industrial-friendly PyTorch implementation of YOLOv5[31]. And the continuous innovation of YOLOv6-v8 in areas such as backbone networks, label allocation, and anchor-free detection. In the recent stage of integrating cutting-edge technologies, the latest deep learning achievements are deeply integrated, such as the NMS-free training strategy of YOLOv10, the lightweight deployment of YOLOv11, especially YOLOv12, which serves as the baseline model of this study[32]. YOLOv12 maintains a lightweight architecture while optimizing the feature extraction module[33], effectively balancing computational efficiency and the multi-scale feature extraction capability required for small target detection in drone scenarios[34]. Its adaptive anchor boxes and dynamic feature fusion mechanism can better adapt to the randomness of target distribution and posture in drone scenes. Compared with YOLOv11[35], YOLO-NAS[36], and RT-DETR[37], YOLOv12 eliminates the shortcomings of insufficient small target detection accuracy, excessive computational overhead, and poor real-time performance. When it is released in 2025, it will integrate attention mechanisms on a large scale for the first time, optimizing spatial modeling capabilities and computational efficiency while maintaining real-time performance[38]. Despite its powerful performance, there is still room for optimization in its general design when dealing with the small targets, geometric distortions and complex backgrounds specific to UAV scenarios, which constitutes the starting point of this study.

Small target detection technology

Small object detection, as a recognized challenge in the field of object detection, is particularly prominent in scenarios such as UAV aerial photography. The challenge lies in their low pixel coverage, sparse features, and susceptibility to noise interference within images[39]. To address this, existing research improvement strategies primarily focus on three directions. The first one is multi-scale feature representation and fusion. The Feature Pyramid Network (FPN) proposed by Lin et al.[3] represents foundational work in this area, achieving top-down path fusion to integrate high-level semantic features with low-level detail features. Subsequently, architectures such as PANet[40], BiFPN[41], and NAS-FPN[42] have emerged successively. These further enhanced the efficiency of cross-scale feature interaction by employing more complex connection mechanisms or automated neural network search techniques. The second is to enhance the utilization of context information[43]. Spatial pyramid pooling captures rich context through multi-scale pooling, while dilated convolution aggregates broader background features by expanding the receptive field, effectively compensating for the insufficiency of the features of small targets themselves[44]. The third approach is data augmentation and generative methods[45]. To address the issue of insufficient small target samples, researchers have adopted techniques such as oversampling and copy-paste to expand the training data[46]. Further, they have utilized generative adversarial networks (GAN) to perform super-resolution reconstruction on small targets, converting low-resolution features into high-resolution forms, thereby transforming them into more manageable large target detection problems. These strategies jointly aim to enhance the model's perception ability and detection accuracy for small targets[47]. Recently, transformer-based detectors have advanced small object detection through innovations such as density-guided query selection for crowded scenes[18] and contrast-aware feature enhancement with uncertainty optimization[17], but complex transformer structures lead to relatively high computational complexity, which limits their application in UAV scenarios with strict real-time and lightweight requirements.

Methodology

Figure 1 shows the overall architecture of the MFR-YOLO model proposed in this paper. The MFR-YOLO architecture mainly consists of three parts: the Backbone, the Neck, and the Head.

Backbone: as the core of feature extraction, the improved backbone network focuses on enhancing multi-scale feature retention, spatial deformation adaptability, and key feature focusing capabilities. Its structural design includes multiple collaborative modules.

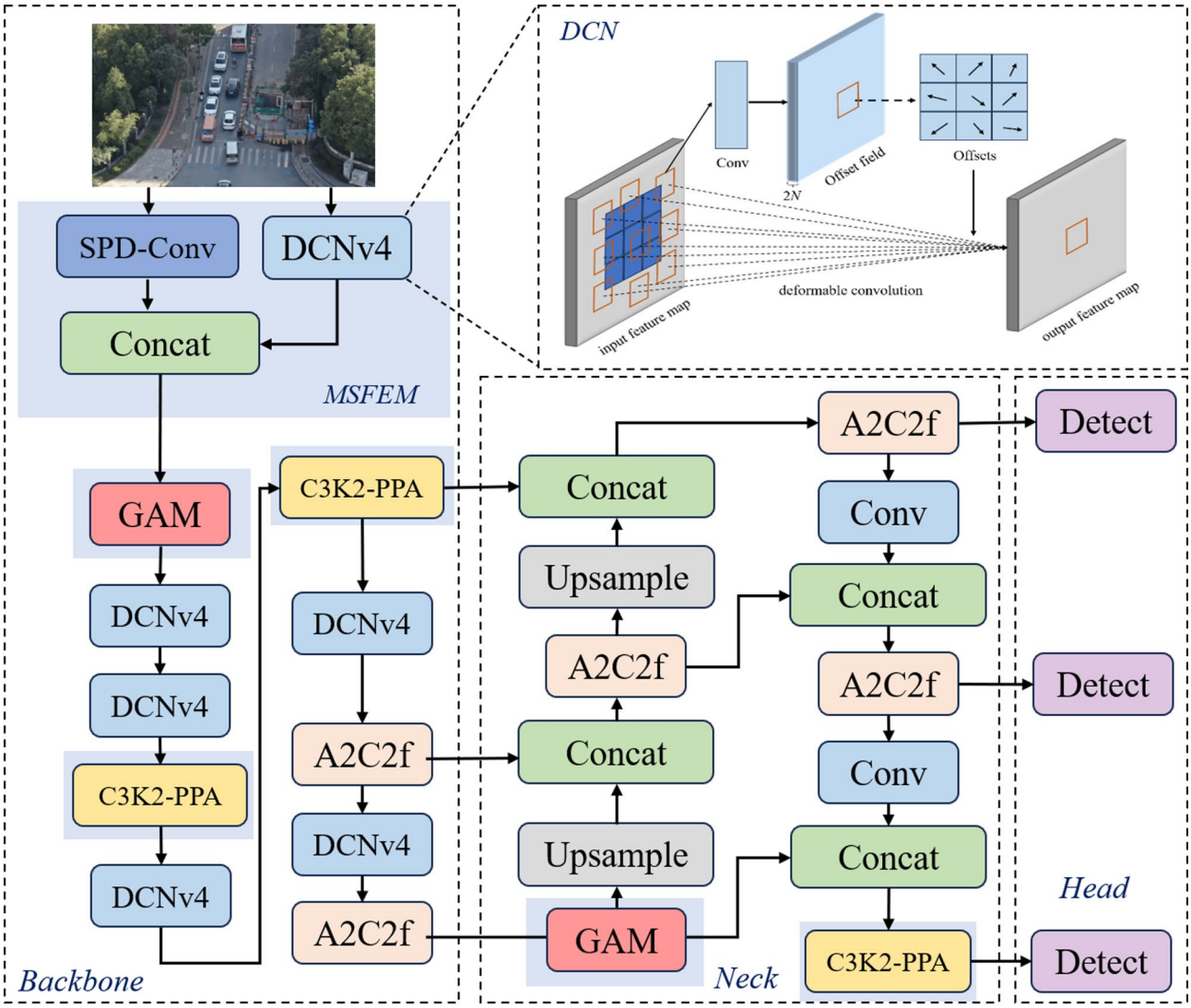


Fig. 1. Overall structure of MFR-YOLO.

(1) MSFEM adopts a dual-branch parallel architecture, which effectively addresses the problem of easy loss of small-target features and enhances the adaptability to spatial deformations such as object rotation, scaling, and distortion. Thus, it can capture more flexible geometric features. The features of the two branches are aggregated through a fusion layer, and then a multi-scale feature map with both detail retention and deformation adaptation characteristics is output.
(2) After the multi-scale feature extraction module of the backbone network, a global attention module with a dual-branch is inserted, which effectively suppresses irrelevant background information and filters out feature information strongly related to the target.
(3) All traditional convolutional modules in the original YOLOv12 backbone network are replaced by DCNv4. This ensures that during the feature transfer process from the input layer to the deep layers, detail features and spatial deformation features will not be lost due to over-compression, improving the integrity of feature transfer.
(4) The original C3K2 module is optimized by integrating it with the Pyramid Pooling Attention (PPA) module to form an enhanced C3K2-PPA module, which effectively improves the network's ability to capture features of targets at different scales.

Neck network: while retaining the multi-scale fusion structure of the original network, a global attention module is inserted at the fusion node of the bottom-layer features. Its function is the same as that of the attention module in the backbone network. By screening the aggregated features again in the spatial and channel dimensions, redundant information is reduced and the transfer efficiency of key features is enhanced. The C3K2-PPA module is used to improve the Neck network's ability to capture features of targets at different scales, from small objects to large objects.

Head network: the head network follows the simplified multi-scale detection path and the dedicated loss function of the original YOLOv12. Due to the improvement in the quality of upstream features, the prediction accuracy and positioning accuracy are further enhanced.

The object-detection process of MFR-YOLO can be divided into five key stages: (a) First, the image is resized to the input size of the model, such as 640 × 640, and the pixel values are mapped to the range of 0–1 through normalization. (b) Subsequently, it enters the feature-extraction stage of the backbone network. The pre-processed image is first input into the multi-scale feature-extraction module. In this module, the SPD-Conv branch retains detailed features, and the DCNv4 branch captures spatial deformation features. After fusion, initial multi-scale features are formed. These features are transferred through the convolutional layers replaced by DCNv4, which avoids the feature compression of traditional convolutions, ensuring that detailed and deformation features are completely transferred to deeper layers. Moreover, at each feature level, the global attention module screens the features in both spatial and channel dimensions, suppressing background noise and focusing on target features. Finally, an optimized multi-scale feature map is output. (c) Next comes the feature aggregation and refinement in the neck network. The multi-scale features output from the backbone enter the neck and are cross-scale fused through the FPN (Feature Pyramid Network) structure. During the fusion process, the global attention module screens the features again to enhance key information. The fused features are then processed by modules such as C3K2-PPA, obtaining more representative multi-scale feature maps that provide high-quality input for the detection head. (d) The head network receives the feature maps output from the neck, generates bounding-box coordinates, classes, and object confidence scores, thus obtaining the initial prediction results. (e) After filtering out duplicate or low-confidence boxes through non-maximum suppression (NMS), the position, class, and confidence of the objects are finally output.

Spatial pyramid depthwise convolution

SPD–Conv is a novel convolutional module optimized for low-resolution image and small-target detection. Figure 2 shows its structure. It aims to replace the strided convolution and pooling layers in traditional CNNs to address the problem of easy loss of features in low-resolution images and small targets. Its core design concept is to enhance the adaptability to spatial deformations while avoiding convolutions with a stride greater than 1 and pooling operations in traditional convolutions, thus better retaining the detailed features of the image. This module improves the ability to capture small and deformed targets while reducing feature loss by separating the convolutional calculations in the spatial dimension and dynamically adjusting the position of the convolutional kernel. It is particularly suitable for object-detection scenarios from the UAV perspective.

Its calculation process can be divided into three key steps:

a. Spatial separation convolution.

A traditional $K \times K$ convolution kernel is decomposed into two parallel $1 \times K$ and $K \times 1$ convolution kernels, which perform feature extraction along the horizontal and vertical directions respectively. The results are then fused through element-wise addition. Given an input feature map $X \in \mathbb{R}^{H \times W \times C_{in}}$, where H and W are the height and width, and C_{in} is the number of input channels. The computation of the separated convolution is:

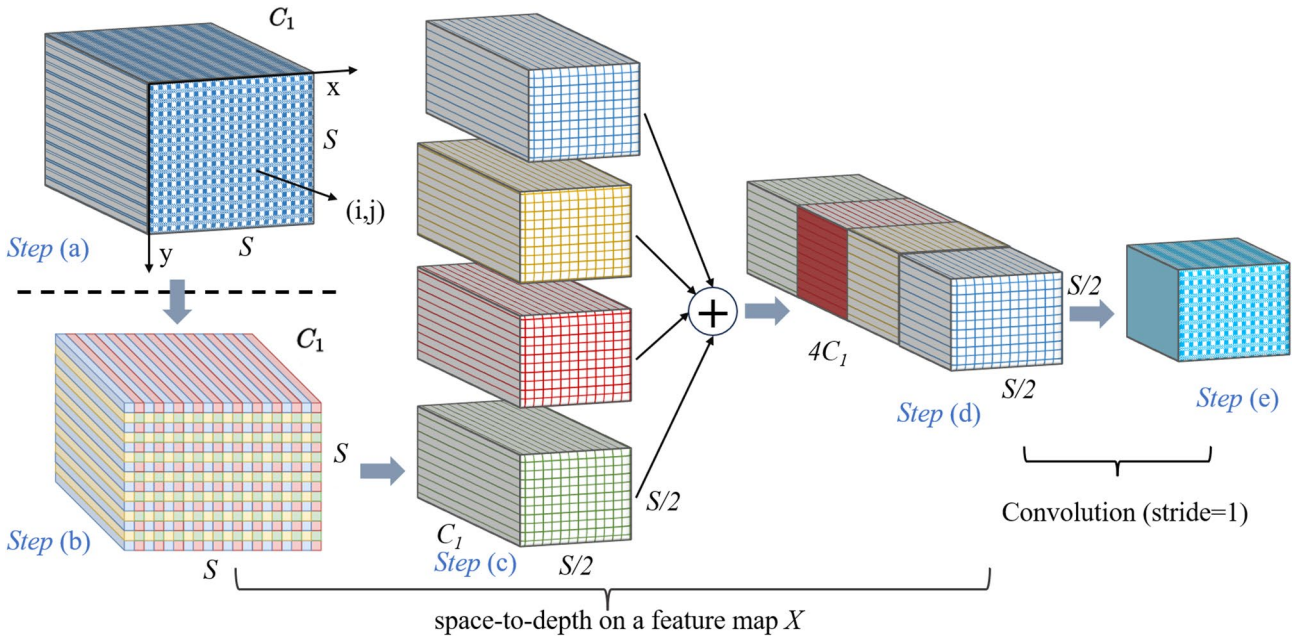


Fig. 2. SPD-Conv module. where (**b-d**) is SPD and (**e**) is Conv.

$$X_{sep} = \mathrm{Conv}_{1\times K}(X) + \mathrm{Conv}_{K\times 1}(X) \tag{1}$$

here, $\mathrm{Conv}_{a\times b}(\cdot)$ denotes a convolution operation with a size of *a*×*b*, and the stride is set to 1 to avoid feature compression. The output $X_{sep} \in \mathbb{R}^{H\times W\times C_{mid}}$, where C_{mid} is the number of intermediate channels.

b. Deformable offset learning.

For the feature map X_{sep} after separated convolution, a lightweight convolution layer 3×3 is used to learn the offset field $\Delta p \in \mathbb{R}^{H\times W\times 2K^2}$, where $2K^2$ represents the (x, y) direction offsets for each position in the K×K convolution kernel. The offset calculation is:

$$\Delta p = \mathrm{Conv}_{3\times 3}(X_{sep}) \tag{2}$$

c. Dynamic sampling and feature aggregation.

Based on the learned offsets Δp, dynamic sampling is performed on the input feature map *X*, followed by feature aggregation using a *K*×*K* convolution. For each position (*i*,*j*) on the output feature map, the corresponding input feature sampling position is:

$$p_{i,j,k} = (i + k_h + \Delta p_{i,j,2k}, j + k_w + \Delta p_{i,j,2k+1}) \tag{3}$$

where $k = 0, 1, ..., K^2 - 1$, and (k_h, k_w) are the relative coordinates within the convolution kernel. The computation of the final output feature map $Y \in \mathbb{R}^{H\times W\times C_{out}}$ is:

$$Y_{i,j} = \sum_{k=0}^{K^2-1} w_k \cdot X(p_{i,j,k}) \tag{4}$$

where C_{out} is the number of output channels, w_k is the weight of the convolution kernel, and $X(p_{i,j,k})$ represents the feature value at the offset position $p_{i,j,k}$ obtained through bilinear interpolation.

Deformable convolution v4

DCNv4 represents the latest iteration of deformable convolutional networks, achieving significant performance breakthroughs through optimization of dynamic aggregation weights and enhancement of memory access efficiency. Its core improvements include: the removal of softmax normalization in spatial aggregation to enhance dynamic expressive capability, and the optimization of memory access patterns to reduce redundant operations, resulting in over a 3x speedup in forward propagation and significantly accelerated convergence. Figure 3 shows several forms of deformable convolution.

a. Input features and initialization.

Assume the input feature map is $X \in \mathbb{R}^{H\times W\times C}$, where *H*, *W*, *C* denote height, width, and the number of channels, respectively. DCNv4 processes features using a group-wise strategy, with each group containing $D = C/G$ channels, where *G* is the number of groups.

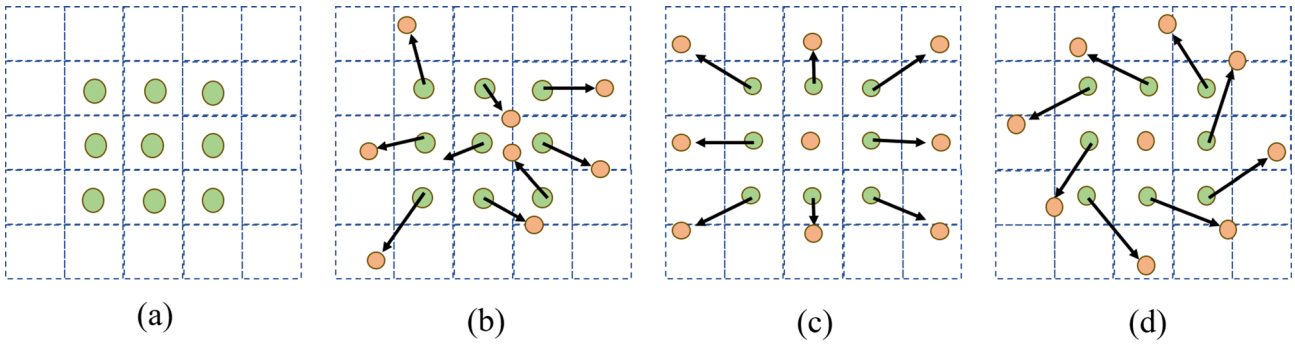


Fig. 3. Different adoption points of deformable convolution. (**a**) is the common sampling method of 3×3 convolution kernels; (**b**) is the sampling deformable convolution, and the change of sampling points after adding the offset; (**c,d**) are special forms of deformable convolution.

b. Dynamic offset generation.

A 3 × 3 convolutional layer processes the input feature X to generate spatial offsets $\Delta P \in \mathbb{R}^{H \times W \times 2K^2}$, where K is the size of the sampling window. The offset calculation formula is:

$$\Delta P = \mathrm{Conv}_{3\times 3}(X) \tag{5}$$

Each offset corresponds to adjustments in the (x,y) coordinates of K×K sampling points.

c. Unbounded dynamic weight aggregation.

For each output position (i, j), DCNv4 samples input features using a dynamic window. Unlike DCNv3, the aggregation weights m_{gk} in DCNv4 are no longer constrained by *softmax*. Instead, unbounded dynamic weights are directly generated via a lightweight network:

$$y_g(i,j) = \sum_{k=1}^{K^2} m_{gk}(i,j) \cdot X_g\left(p_0 + p_k + \Delta p_{gk}(i,j)\right) \tag{6}$$

where p_0 is the reference coordinate of the output position, and p_k is predefined relative coordinates of sampling points. $\Delta p_{gk}(i,j)$ is the offset of the k-*th* sampling point in the g-*th* group. $m_{gk}(i,j)$ is unbounded dynamic aggregation weights generated by a lightweight network.

d. Thread merging and bilinear interpolation sharing.

The D channels within the same group share identical sampling offsets and aggregation weights. DCNv4 merges computations across multiple channels into a single thread, avoiding redundant reads of the same offsets and weights. Simultaneously, bilinear interpolation coefficients are computed once and reused across all channels, reducing memory access overhead by a factor of D.

e. Concatenation of output features.

Output features y_g from each group are concatenated along the channel dimension to form the final output $Y \in \mathbb{R}^{H \times W \times C}$:

$$Y = \mathrm{concat}(y_1, y_2, \ldots, y_G, \mathrm{axis} = 1) \tag{7}$$

Global attention mechanism

Figure 4 shows GAM, a dual-branch feature enhancement mechanism that combines spatial attention and channel attention. It aims to focus on key target features and suppress background noise through dynamic weight allocation, making it suitable for object detection tasks in complex scenes from the perspective of unmanned aerial vehicles. The spatial attention branch calculates the correlation between pixels through a self-attention

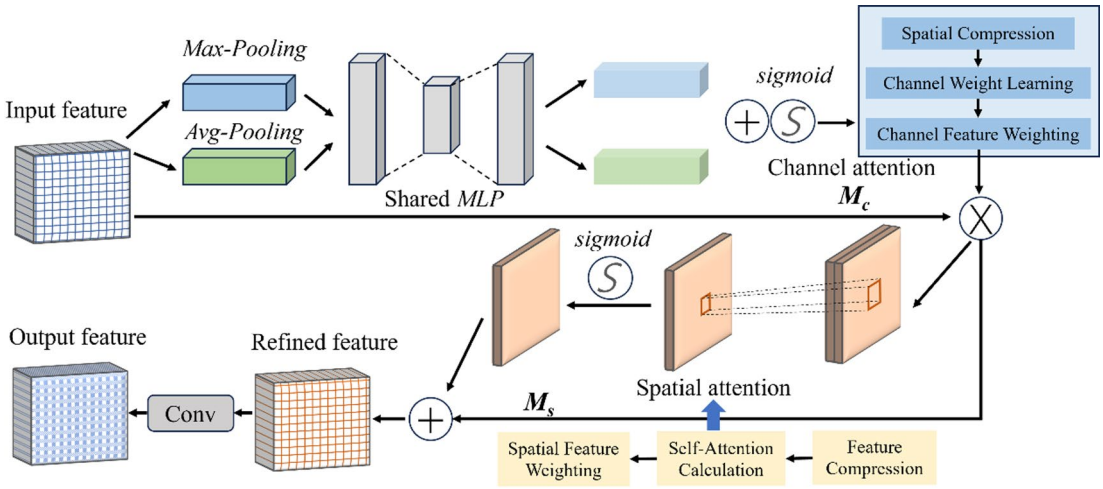


Fig. 4. The structure of GAM.

mechanism, focusing on the contour areas of targets such as vehicles and pedestrians and suppressing irrelevant backgrounds like the sky and ground. The channel attention branch, on the other hand, learns the weights of feature channels to select channels strongly related to the target, such as edges and textures, and weaken background noise channels like blurred shadows, further enhancing the discriminability of features.

a. Spatial attention branch.

This branch models the correlation between pixels through a self-attention mechanism to enhance the weights of target regions and suppress the background, the process involves three steps:

Feature compression: perform global average pooling on the input feature $X_{spatial}$ along the channel dimension to obtain a spatial feature map $S \in \mathbb{R}^{H \times W \times 1}$:

$$S(i,j) = \frac{1}{C} \sum_{c=1}^{C} X_{spatial}(i,j,c) \tag{8}$$

Self-attention calculation: learn spatial correlation weights through a lightweight convolutional layer to generate a spatial attention map $M_s \in \mathbb{R}^{H \times W \times 1}$, where each pixel value represents the importance of the corresponding position:

$$M_s = \sigma\left(\mathrm{Conv}_{7\times 7}(S)\right) \tag{9}$$

where σ is the Sigmoid activation function, with outputs in the range [0, 1].

Spatial feature weighting: element-wise multiply the attention map M_s with the input feature X to strengthen target region features:

$$X_s = X \odot M_s \tag{10}$$

where $\odot$denotes element-wise multiplication.

b. Channel attention branch.

This branch filters target-related feature channels and weakens noise channels by learning channel weights, the computation proceeds as follows:

Spatial compression: perform global average pooling on the input feature $X_{channel}$ along the spatial dimension to obtain a channel feature vector $C_{avg} \in \mathbb{R}^{1\times 1\times C}$:

$$C_{avg}(c) = \frac{1}{H \times W} \sum_{i=1}^{H} \sum_{j=1}^{W} X_{channel}(i,j,c) \tag{11}$$

Channel weight learning: leverage a bottleneck structure to learn the importance weights of channels. This structure first applies a 1×1 convolution for dimensionality reduction, followed by a ReLU activation function, and then another 1×1 convolution for dimensionality elevation. Through this process, a channel attention vector $M_c \in \mathbb{R}^{1\times 1\times C}$ is generated, encapsulating the significance of each channel:

$$M_c = \sigma\left(\mathrm{Conv}_{1\times 1}\left(\mathrm{ReLU}\left(\mathrm{Conv}_{1\times 1}(C_{avg})\right)\right)\right) \tag{12}$$

Channel feature weighting: element-wise multiply the attention vector M_c with the spatially weighted feature X_s to highlight target-related channels:

$$X_c = X_s \odot M_c \tag{13}$$

C3K2-PPA module

The C3K2-PPA module is an enhanced feature processing unit formed by integrating the PPA module into the original C3K2 module of YOLOv12. Figure 5 illustrates the complete architecture and computational workflow. Its core function is to improve the capability of capturing features of targets at different scales through multi-branch collaborative learning, making it particularly suitable for drone-view scenarios where small targets, distorted targets, and complex backgrounds coexist. The design philosophy retains the advantages of cross-stage local feature fusion in the C3K2 module while expanding the range of feature receptive fields through the parallel branch structure of the PPA module, enabling multi-dimensional feature modeling from local details to global correlations.

In C3K2-PPA module, the input feature X first passes through a 1×1 convolution for channel adjustment. The resulting feature X_{c3} is then processed by three parallel branches: (1) Local Branch with 3×3 convolution for

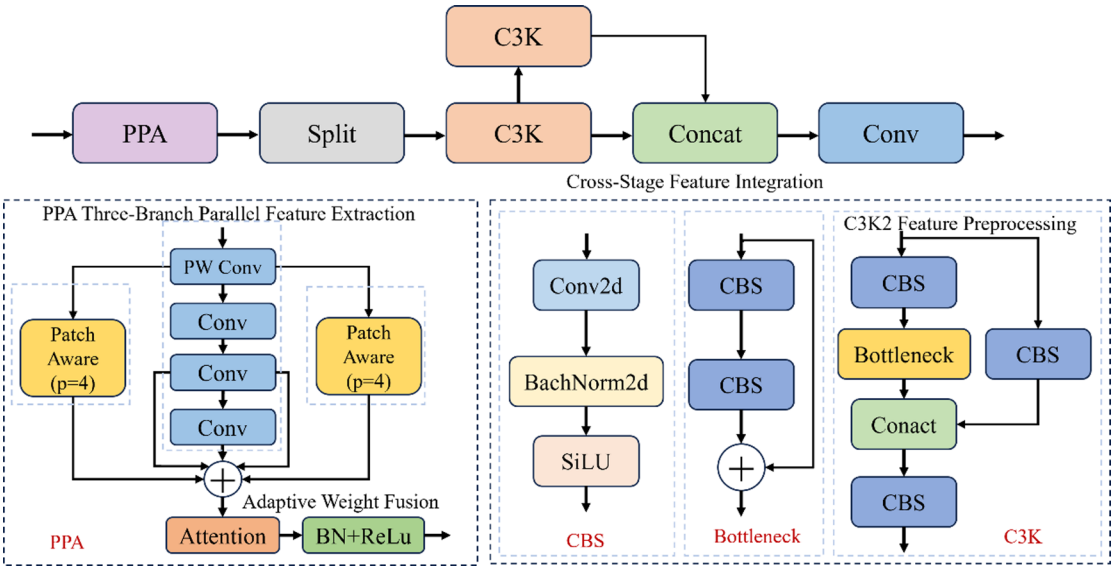


Fig. 5. C3K2-PPA module.

fine detail preservation. (2) Global Branch with 6 × 6 convolution for scene context modeling. (3) Serial Branch with cascaded 1 × 1→3 × 3→1 × 1 convolutions for multi-scale aggregation. The branch outputs are unified to the same channel dimension via 1 × 1 convolutions and adaptively fused using learnable weights α, β, γ. Finally, the fused feature is added to the residual connection from the input to produce the output X_{out}.

Assuming the input feature map is $X \in \mathbb{R}^{H \times W \times C}$, where H and W are the height and width of the feature map, and C is the number of channels. The computational flow of the C3K2-PPA module is divided into the following steps:

a. C3K2 basic feature preprocessing.

The input feature X passes through the initial convolution layer of the C3K2 module for channel adjustment, outputting an intermediate feature $X_{c3} \in \mathbb{R}^{H \times W \times C'}$, where C' is the adjusted number of channels, to provide a consistent input for subsequent PPA branch processing:

$$X_{c3} = \text{Conv}_{1\times1}(X) \tag{14}$$

b. PPA three-branch parallel feature extraction.

Local branch: this branch uses a single 3 × 3 convolution with padding 1 to maintain spatial dimensions, focusing on local detailed features such as target edges and textures. By limiting the coverage of the convolution kernel, interference from background noise is reduced:

$$X_{local} = \text{Conv}_{k1\times k1}(X_{c3}) \quad (k1 = 2 \text{ or } 3) \tag{15}$$

Global branch: employs convolution or pooling operations with a large patch size to capture global correlations between non-overlapping regions. To avoid excessive computational load, channel compression is usually performed first via 1 × 1 convolution:

$$X_{global} = \text{Conv}_{1\times1}\left(\text{Conv}_{k2\times k2}(X_{c3})\right) \quad (k2 = 6 \text{ or } 8) \tag{16}$$

Serial branch: adopts a cascaded structure of '1 × 1 → 3 × 3 → 1 × 1' convolution to gradually expand the receptive field and aggregate multi-scale semantic features. The 1 × 1 convolution is used for channel dimension reduction and elevation, while the 3 × 3 convolution is responsible for cross-scale feature fusion:

$$X_{serial} = \text{Conv}_{1\times1}\left(\text{Conv}_{3\times3}\left(\text{Conv}_{1\times1}(X_{c3})\right)\right) \tag{17}$$

c. Adaptive weight fusion.

The output features of the three branches, X_{local}, X_{global} and X_{serial}, first undergo 1 × 1 convolution to unify their channel numbers to C'. Then, they are adaptively fused using learnable weight parameters α, β, γ to highlight high-contribution features:

$$X_{fusion} = \alpha \cdot X_{local} + \beta \cdot X_{global} + \gamma \cdot X_{serial} \tag{18}$$

Among them, the weight parameters are dynamically learned through backpropagation, with the formula:

$$\alpha = \frac{\exp(w_1)}{\exp(w_1) + \exp(w_2) + \exp(w_3)}, \quad \beta = \frac{\exp(w_2)}{\sum \exp(w_i)}, \quad \gamma = \frac{\exp(w_3)}{\sum \exp(w_i)} \tag{19}$$

where w_1, w_2, w_3 are the weight parameters to be learned.

Cross-stage feature integration

The fused feature X_{fusion} is added to the residual connection of the C3K2 module, outputting the final feature $Y \in \mathbb{R}^{H \times W \times C}$:

$$Y = X_{fusion} + \mathrm{Conv}_{1\times1}(X) \tag{20}$$

The C3K2-PPA module achieves a stepwise expansion of feature receptive fields through its multi-branch structure, the local branch retains details of small targets, the global branch models scene context, and the serial branch balances semantic levels. Meanwhile, the adaptive weight fusion mechanism enables the model to dynamically adjust the contribution of each branch according to input content, demonstrating stronger feature adaptability in multi-scale target detection tasks under drone views.

Experiments

Experiment settings

The experiment was completed on a high-performance server with an Intel(R) Xeon(R) Silver 4210R CPU and 256 GB DDR4-3200 memory. The GPU adopts 6×GeForce GTX 3090. The system uses Ubuntu 22.04 LTS and is equipped with CUDA 12.2, cuDNN 8.9.4 and PyTorch 2.1.0. Table 1 shows the relevant parameters of the MFR-YOLO model in the experiment.

Datasets

The mainstream public dataset VisDrone 2021 for unmanned aerial vehicle scenarios and the vehicle detection dataset UA-DETRAC were selected to cover different aerial photography scenarios and target characteristics.

VisDrone2021[48]: the dataset consists of 400 video clips, 265,228 frames and 10,209 static aerial images, and we selected the 10,209 static aerial images for our experiments. The scenes cover urban main roads, rural roads, squares, Bridges, parking lots, etc., and span various weather and lighting conditions such as day and night, sunny/rainy/foggy. The image resolution ranges from 640 × 640 to 2000 × 1500, providing over 2.6 million fine horizontal bounding box annotations for 10 common types of targets: pedestrians, people, bicycles, cars, vans, trucks, tricycles, sunshade tricycles, buses, motorcycles, along with attributes such as occlusion, truncation, and visibility. It is applicable to research on small target detection, anti-interference against complex backgrounds, etc.

UA-DETRAC[49]: the dataset contains 82,000 key frames extracted from 10 h of continuous aerial video footage, covering over 60 typical traffic scenarios with an image resolution of 960 × 540. It provides 1.4 million annotated vehicle instances across three primary categories including automobiles, buses, and vans, along with 14 attribute labels describing factors such as vehicle size, occlusion, weather, and lighting conditions. Although originally designed for multi-object tracking, we adapt it for object detection by sampling one frame every five frames, which yields 16,400 frames for our experiments. These frames are randomly split into training and testing sets with a 70% and 30% ratio respectively, and the original category labels are mapped to our task-specific categories to ensure consistency with the detection requirements of our study.

Evaluation indicators

mAP50: it represents the average accuracy when the IoU (intersection and union ratio) threshold is 0.5.

$$\mathrm{AP} = \int_0^1 P(R)dR \quad \Rightarrow \quad \mathrm{mAP} = \frac{1}{N}\sum_{i=1}^{N} \mathrm{AP}_i \tag{21}$$

Parameter	Description	Value
Input size	Input resolution	640×640
Batch size	Batch size	64
Initial lr	Initial learning rate	1e-3
Augment level	Enhance the intensity of the training data	1.0
Weight decay	L2 regularization coefficient	5e-4
Epoch	Number of training iterations	100

Table 1. Parameter Settings of the MFR-YOLO model.

Here, $P(R)$ is the area under the Precision-Recall curve, N is the number of target categories, and AP_i is the average accuracy of the i-th target.

mAP50-small: a specialized metric designed for common small targets (targets with a resolution of less than 32×32 pixels) in drone scenarios.

$$\mathrm{mAP}_{\mathrm{Small}} = \frac{1}{N_{\mathrm{Small}}} \sum_{i=1}^{N_{\mathrm{Small}}} \mathrm{AP}_i \tag{22}$$

Miss rate: a specialized metric designed for common small targets in drone scenarios, specifically targets with a resolution of less than 32×32 pixels.

$$\text{Miss Rate} = 1 - \text{Recall} = 1 - \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}} \tag{23}$$

FPS (frame rate): it measures the inference speed of a model, that is, the number of image frames that can be processed per second.

$$\mathrm{FPS} = \frac{\text{Batch Size}}{\text{Total Inference Time}} \tag{24}$$

Comparative experiments

To verify the overall superiority of MFR-YOLO, baseline models from the YOLO series and mainstream models from other frameworks, were selected for comparison. Performance evaluations were carried out on the VisDrone2021 and UA-DETRAC datasets. The introductions of each baseline model are as follows:

Faster R-CNN[50]: a two-stage detection architecture that achieves automatic candidate region generation by introducing a Region Proposal Network (RPN). It combines a ResNet backbone with ROI pooling operations, delivering strong detection accuracy in complex environments but facing limitations in inference speed.

YOLOv5[31]: a lightweight single-stage detection model. This architecture utilizes pre-trained weight initialization combined with adaptive training strategies. By incorporating mosaic data augmentation and adaptive anchor calculation techniques, it achieves strong deployment capabilities even on edge computing devices.

YOLOv8[51]: this detection framework employs an anchor-free design, integrating the CSPDarknet53 backbone with a task-aligned detection head. It simultaneously supports multiple tasks including object detection, classification, and instance segmentation, achieving an effective balance between accuracy and inference speed.

YOLOv10[52]: this model employs a dual-branch architecture that decouples classification and regression tasks. Combined with a global context awareness mechanism and dynamic label allocation strategy, it further enhances the model's adaptability to complex scenes.

YOLOv12[53]: by introducing dynamic convolution operators and multi-level feature fusion mechanisms, this model significantly enhances small object detection performance while maintaining high inference efficiency. Its optimized feature transfer strategy also helps reduce recognition errors in complex scenarios such as image distortion.

RetinaNet[54]: a one-stage focal detection network. It solves the problem of positive-negative sample imbalance through Focal Loss. Combining with the Feature Pyramid Network (FPN), it realizes multi-scale detection, improving the accuracy of small targets while maintaining a relatively fast speed.

SD-YOLO[55]: a small-object detection specialized model optimized for small target feature extraction, which enhances the detection ability of small objects by designing a dedicated feature enhancement module.

DINO-X[56]: a recent transformer-based detector with strong global feature modeling capabilities, which improves target detection accuracy by optimizing the query selection mechanism.

DDRN[57]: an AV-specific detection method, which is optimized for complex scene adaptation and has good performance in target detection under dynamic scenarios.

MFR-YOLO (ours): the improved YOLOv12 method proposed in this paper. It newly adds a multi-scale feature extraction module MSFEM, introduces a global attention module GAM, and improves the C3K2 module by introducing PPA.

The evaluation results of each model on the VisDrone2021 and UA-DETRAC datasets are shown in Tables 2 and 3.

On the two major UAV scene datasets, VisDrone2021 and UA-DETRAC, the overall detection accuracy of MFR-YOLO is significantly superior to that of the comparison models, verifying the effectiveness of the improvement strategy, Fig. 6 shows an intuitive comparison result. On the VisDrone2021 dataset, the mAP50 of MFR-YOLO reaches 40.1%, an increase of 1.8% compared to the baseline model YOLOv12. It also significantly exceeds mainstream models such as YOLOv10 and RetinaNet, and outperforms specialized detectors including SD-YOLO, DDRN, and the Transformer-based DINO-X, performing the best among all the comparison models. On the UA-DETRAC dataset, the advantage of MFR-YOLO is even more obvious. Its mAP50 reaches 54.6%, an increase of 2.9% compared to YOLOv12, far exceeding YOLOv8 and Faster R-CNN. This result indicates that MFR-YOLO effectively enhances the overall recognition ability of targets in complex aerial photography scenes through multi-module collaborative optimization.

In addition, on the VisDrone2021 dataset, the mAP50-Small of MFR-YOLO reaches 31.1%, an increase of 2.7% compared to YOLOv12 and 5.5% compared to YOLOv10, and it also outperforms the small-object specialized model SD-YOLO, whose mAP50-Small is 29.2%. On the UA-DETRAC dataset, the mAP50-Small

Model	mAP50(%)	mAP50-small (%)	Miss rate (%)	FPS
Faster R-CNN	31.2	18.7	17.3	188
YOLOv5	31.7	21.2	15.3	176
YOLOv8	33.6	22.3	13.5	165
YOLOv10	34.2	25.6	12.9	151
RetinaNet	37.4	27.8	10.7	146
SD-YOLO	37.8	29.2	11.2	115
DINO-X	39.5	28.9	10.5	98
DDRN	36.7	26.8	12.1	122
YOLOv12	38.3	28.4	10.9	140
MFR-YOLO	**40.1**	**31.1**	**9.4**	**128**

Table 2. Performance comparison of different models on VisDrone2021. Significant values are in bold.

Model	mAP50(%)	mAP50-small(%)	Miss rate(%)	FPS
Faster R-CNN	38.6	20.6	15.2	194
YOLOv5	40.2	24.3	14.4	183
YOLOv8	42.5	26.0	12.8	172
YOLOv10	45.3	27.5	12.2	166
RetinaNet	48.4	29.1	11.7	151
SD-YOLO	50.2	30.1	10.8	109
DINO-X	52.3	29.8	10.3	92
DDRN	49.5	28.2	11.5	117
YOLOv12	51.7	29.4	11.1	142
MFR-YOLO	54.6	32.3	9.2	130

Table 3. Performance comparison of different models on UA-DETRAC.

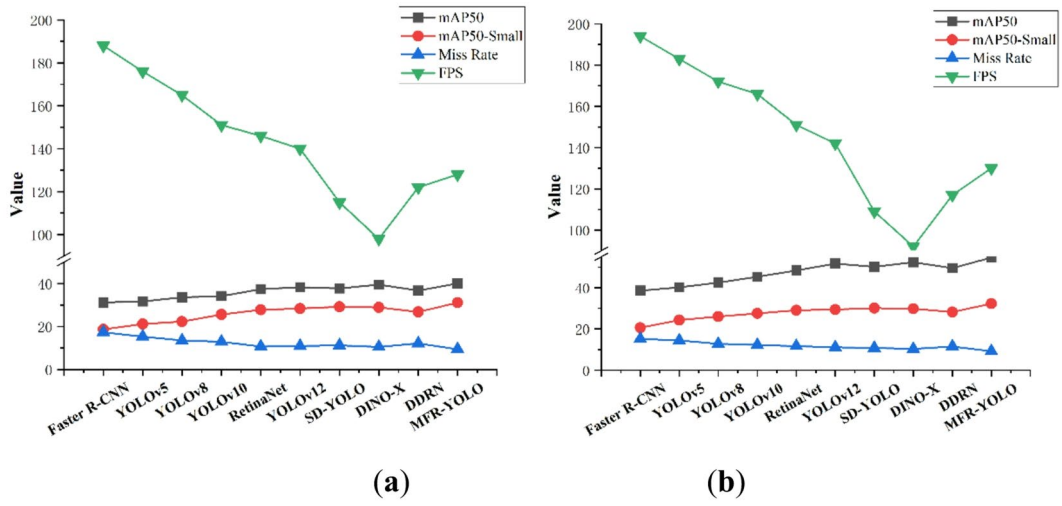


Fig. 6. Comparison of experimental results of different models. (**a**) result on VisDrone2021; (**b**) result on UA-DETRAC.

reaches 32.3%, an increase of 2.9% compared to YOLOv12 and 3.2% compared to RetinaNet. This result demonstrates the advantage of MFR-YOLO in small-target detection performance. The main reason is that the newly added multi-scale feature extraction module retains high-resolution details through SPD-Conv and dynamically adapts to the geometric deformation of small targets in combination with DCNv4, significantly enhancing the ability to capture small-scale targets.

Miss Rate, as the core metric for measuring the comprehensiveness of model target recognition, MFR-YOLO still performs outstandingly in this metric. The miss rate on the VisDrone2021 dataset was 9.4%, which was 1.5% and 1.3% lower than that of YOLOv12 and RetinaNet respectively, and also lower than the miss rates of SD-YOLO,

model	Parameter count (M)	Peak memory usage (GB)	FLOPs (G)	Inference time (ms/frame)
Faster R-CNN	42.6	2.8	18.7	5.3
YOLOv5	27.3	1.9	12.5	3.8
YOLOv8	30.1	2.1	14.2	4.1
YOLOv10	32.7	2.3	15.8	4.5
RetinaNet	38.9	2.6	17.3	5
SD-YOLO	35.2	2.4	16.5	6.2
DINO-X	58.7	3.5	28.9	8.7
DDRN	37.8	2.5	17.8	5.8
YOLOv12	34.5	2.2	15.1	4.8
MFR-YOLO	**33.8**	**2.2**	**15.5**	**4.9**

Table 4. Computational efficiency comparison of different models. Significant values are in bold.

Model structure	mAP50(%)	mAP50-small(%)	Miss rate(%)	FPS
YOLOv12	38.3	28.4	10.9	140
+ MSFEM	38.7	28.8	10.7	138
+ MSFEM + DCNv4	38.9	29.4	10.4	135
+ MSFEM + DCNv4 + GAM	39.3	30.3	10.3	131
+ MSFEM + DCNv4 + C3K2	37.4	27.8	11.0	137
+ MSFEM + DCNv4 + PPA	38.1	28.6	10.5	132
+ MSFEM + DCNv4 + C3K2-PPA	38.9	30.7	9.9	130
MFR-YOLO	**40.1**	**31.1**	**9.4**	**128**

Table 5. Results of MFR-YOLO Ablation Experiment. Significant values are in bold.

DINO-X and DDRN, which are 11.2%, 10.5% and 12.1% respectively. The miss rate of the UA-DETRAC dataset has further decreased to 9.2%. In terms of real-time performance, the FPS of MFR-YOLO in VisDrone2021 and UA-DETRAC is 128 and 130 respectively. Although the inference speed is slightly lower than that of YOLOv12, it is still higher than that of models such as RetinaNet and far exceeds the benchmark requirement of 30 FPS for real-time detection. This precision-efficiency balance benefits from the PPA improved C3K2 module, which efficiently aggregates features through local-global-serial branches, enhancing multi-scale context modeling capabilities while avoiding a significant increase in the number of parameters and computational load, ultimately achieving the optimal trade-off between performance and real-time performance.

To further verify the UAV deployment feasibility of MFR-YOLO, we supplemented the evaluation with key computational analysis metrics, including parameter count, peak memory usage, FLOPs, and inference time on the NVIDIA Jetson AGX Orin platform, which is a representative example of typical UAV-embedded hardware. The experiments were conducted under conditions that closely simulate practical UAV deployment: the input image size was set to 640×640 to match the standard resolution of UAV aerial photography. A batch size of 1 was adopted to emulate real-time single-frame inference scenarios, the operating system was Ubuntu 20.04 LTS, and the inference framework was PyTorch 2.4.1 with TensorRT acceleration to ensure consistency with actual deployment configurations.

As shown in Table 4, MFR-YOLO maintains a relatively low parameter count of 33.8 M and 15.5 G FLOPs compared to models such as DINO-X, while its peak memory usage of 2.2 GB remains well within the constraints of UAV hardware and consistent with the baseline model YOLOv12. The inference time of 4.9 ms per frame is slightly longer than that of lightweight models such as YOLOv5 and YOLOv8, yet it is significantly shorter than that of heavier models such as DINO-X and SD-YOLO, fully satisfying the real-time deployment requirements of UAVs. These results further confirm that MFR-YOLO achieves an optimal balance between detection performance and computational efficiency, making it highly suitable for practical UAV deployment.

Ablation experiment

To quantify the contributions of each improved module, taking YOLOv12 as the baseline and adopting the ablation strategy of gradually adding modules", the performance of each configuration of MFR-YOLO was tested on VisDrone2021. The results are shown in Table 5.

The results show that each module has a clear division of labor and synergy in the improvement of accuracy, while achieving a balance between performance and real-time performance.

In terms of the core accuracy indicator mAP50, the model accuracy shows a step -by-step increase after gradually adding modules. When only the MSFEM is added, the mAP50 increases from the baseline of 38.3% to 38.7%, an increase of 0.4%, preliminarily verifying the fundamental role of MSFEM in capturing multi-scale features. After adding DCNv4, the mAP50 further increases to 38.9%, indicating that the deformable convolution's ability to adapt to the geometric deformation of targets can supplement the distortion feature

information not covered by MSFEM. After adding the GAM, the mAP50 breaks through to 39.3%, with a more significant increase than the previous two steps, reflecting the advantage of GAM in suppressing background interference and focusing on key target regions through spatial-channel dual attention. Separate ablations on C3K2-PPA components reveal that adding only C3K2 drops mAP50 to 37.4%, while PPA alone yields 38.1%. Only the complete C3K2-PPA restores performance to 38.9%, demonstrating that their integration is essential. Finally, the fully integrated MFR-YOLO boosted mAP50 to 40.1%, representing a 1.8% increase over the baseline and validating the effectiveness of synergistic optimization across all four modules.

Based on ablation results for small object detection performance mAP50-Small, the contributions of different modules exhibit high specificity. The initial MSFEM module only brings a modest gain of 0.4%. In sharp contrast, after the introduction of the DCNv4 module, this metric increases significantly to 29.4%, which confirms its strong ability to model the geometric distortions of small objects and effectively compensates for the limitations of MSFEM in this regard. Adding C3K2 alone drops small-object accuracy to 27.8%, while PPA alone achieves 28.6%. The complete C3K2-PPA jumps to 30.7%, which represents a 2.3% improvement over the baseline. This result confirms that integration is critical for small-target enhancement. Finally, under the complete model configuration, mAP50-Small reaches 31.1%, a 2.7% increase compared to the baseline. This result clarifies that the functionally complementary combination of MSFEM, DCNv4, and C3K2-PPA is the core for significantly improving small-object detection performance.

The optimization of the Miss Rate mainly depends on the collaboration between GAM and C3K2-PPA. The Miss Rate of the baseline model is 10.9%. After adding GAM, it drops to 10.3%, reflecting the enhancement effect of the attention mechanism on the recognition of sparse and occluded targets. C3K2 alone increases miss rate to 11.0%, which is worse than the baseline, while PPA alone reduces it to 10.5%. The complete C3K2-PPA achieves 9.9%, which is a 1.0% reduction from the baseline. This demonstrates that integrated multi-scale aggregation effectively reduces missed detections. After the full-module integration, the Miss Rate drops to 9.4%, a decrease of 1.5% compared to the baseline, verifying the improvement effect of 'attention screening + multi-scale aggregation' on the overall target recognition ability.

In terms of real time performance FPS, the impact of adding each module on efficiency is controllable. The FPS gradually decreases from the baseline of 140 FPS to 128 FPS for the full module model, with a decrease of only 8.6%, and it is still far higher than the real time detection requirement of UAV, which is no less than 30 FPS. Among them, MSFEM and DCNv4 have a relatively small impact on the FPS, with each reducing 2–3 FPS. Due to the introduction of attention calculation and multi-branch aggregation, GAM and C3K2-PPA reduce the FPS by 4 FPS and 1 FPS respectively, but the overall performance remains efficient, proving that the lightweight design of each module effectively balances the contradiction between accuracy improvement and increased computational cost.

In summary, the improved modules of MFR-YOLO are functionally complementary. MSFEM lays the foundation for multi-scale feature extraction, DCNv4 strengthens the deformation adaptation ability, GAM optimizes the target-background discriminability, and C3K2-PPA, through integration of multi-branch design, significantly improves small-target detection and reduces miss rate. The four modules work together to achieve a comprehensive optimization, providing a clear path for the contribution of each module to the improvement of detection performance in UAV scenarios.

Visual analysis

Two representative images with dense small targets and complex backgrounds were selected from the dataset to compare the detection effects of YOLOv12 and MFR-YOLO, as shown in Fig. 7.

The results show significant differences in the detection capabilities of the two types of models. In the first image ((a) and (b) in Fig. 7.), YOLOv12 detected only 20 targets, and there were obvious missed detections of small-sized pedestrians and non-motor vehicles densely distributed in the distance. In contrast, MFR-YOLO successfully detected 25 targets. By retaining high-resolution details through the MSFEM, it achieved precise capture of small targets. In the second image ((c) and (d) in Fig. 7), the number of targets detected by YOLOv12 was 171. In the scenario of complex building backgrounds and partial occlusion of targets, false detections and missed detections were prominent. However, MFR-YOLO detected 183 targets. Its GAM module effectively suppressed background noise interference. At the same time, combined with the multi-scale context aggregation of the C3K2-PPA module, it enhanced the recognition ability for occluded targets and low-contrast targets. Similarly, in the third and fourth images, the number of targets detected by YOLOv12 was 91 and 69 respectively, while MFR-YOLO detected 102 and 93 respectively. Overall, the visualization results intuitively confirm the superiority of MFR-YOLO in small-target detection and complex-background adaptation, which is consistent with the conclusion of the improvement in mAP50-Small in the quantitative experiments.

Conclusion

This study investigates the technical bottlenecks of UAV scene target detection. To address the core challenges of high proportion of small targets, severe perspective distortion, complex background interference and strict real-time requirements in this scene, an enhanced MFR-YOLO model is proposed with YOLOv12 as the baseline model to achieve the collaborative optimization of accuracy and efficiency. This study proposes a solution through four key improvements: First, it designs a MSFEM module, integrating SPD-Conv and DCNv4, taking into account both high-resolution detail retention and spatial deformation adaptation. The second is to replace the traditional convolution of the backbone network with DCNv4 to enhance the integrity of feature transfer. The third is to embed the GAM, which suppresses background noise through dual screening of channels and space. Fourth, optimize the C3K2-PPA module and enhance the multi-scale target capture capability by relying on multi-branch aggregation.

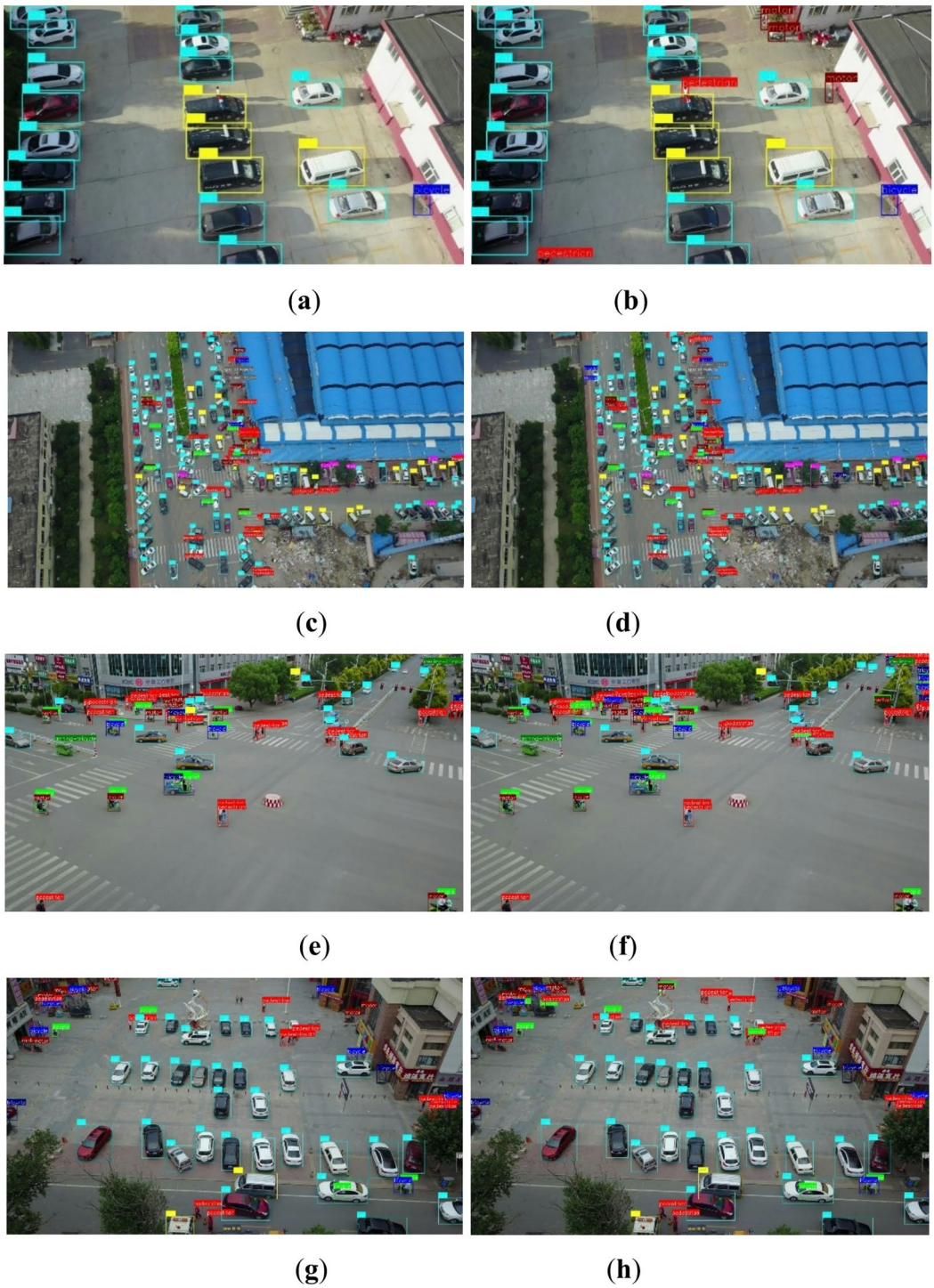


Fig. 7. The detection results of YOLOv12 and MFR-YOLO. (**a**), (**c**), (**e**) and (**g**) are the results of YOLOv12, and (**b**), (**d**), (**f**) and (**h**) are the results of MFR-YOLO.

Experimental results on the VisDrone2021 and UA-DETRAC benchmark datasets demonstrate that the proposed MFR-YOLO model significantly outperforms other comparison algorithms in overall detection accuracy. Notably, the model exhibits outstanding recognition capabilities for small-sized objects, with false negative rates effectively controlled. Although MFR-YOLO's inference speed is slightly slower than YOLOv12, it still comfortably meets real-time requirements for practical applications with a substantial margin, achieving a favorable balance between detection accuracy and computational efficiency. Systematic ablation experiments validate that each integrated improvement module positively contributes to enhancing detection performance from a drone perspective. The proposed technical framework not only offers novel solutions for UAV-based target detection but also provides valuable insights for practical applications such as intelligent traffic management and

disaster emergency response. Furthermore, it establishes a methodological approach for optimizing detection models tailored to specific scenarios.

Data availability

The datasets analyzed in this research are publicly available online: VisDrone2021 ([https://aiskyeye.com/visdrone-2021-leaderboard/](https:/aiskyeye.com/visdrone-2021-leaderboard)), and UA-DETRAC ([http://detrac-db.rit.albany.edu/](http:/detrac-db.rit.albany.edu)). Further inquiries can be directed to the corresponding author.

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Author contributions

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Declarations

Competing interests

The authors declare no competing interests.

Additional information

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